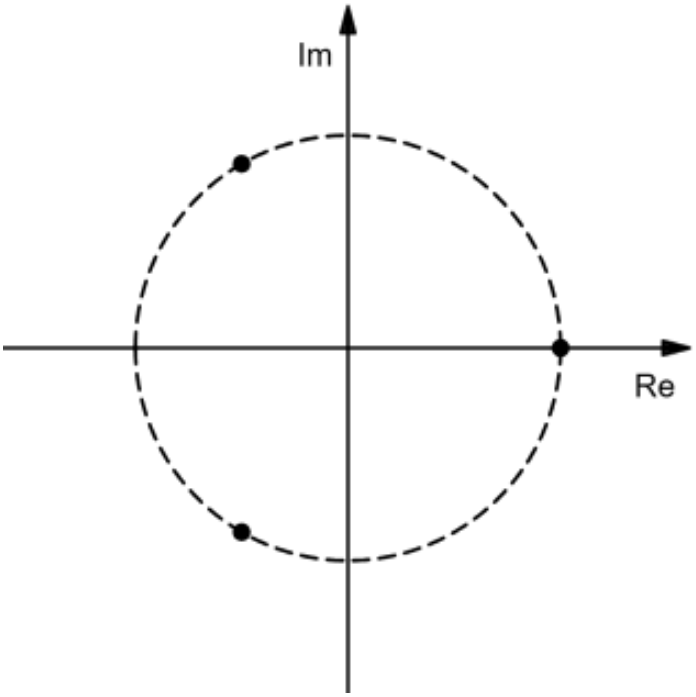
2.07 Complex Numbers: The Geometry of nth Roots

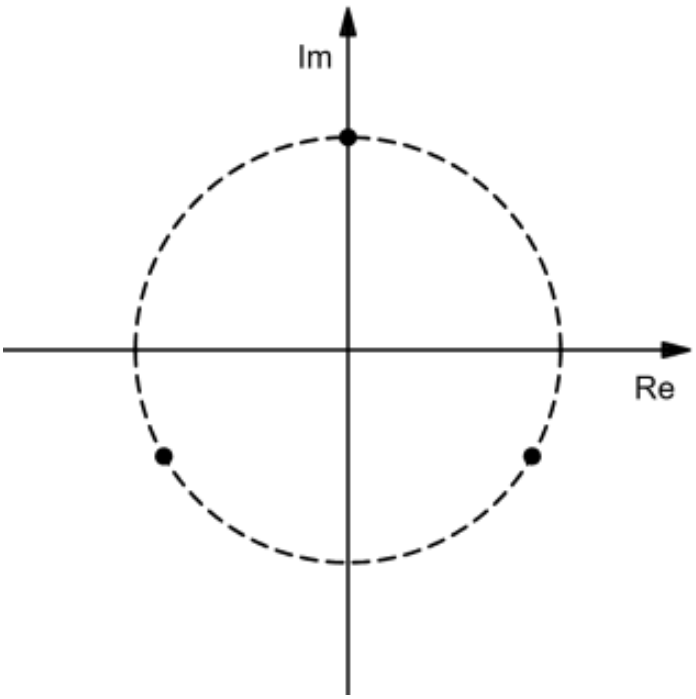
Worked Solutions

1 a)



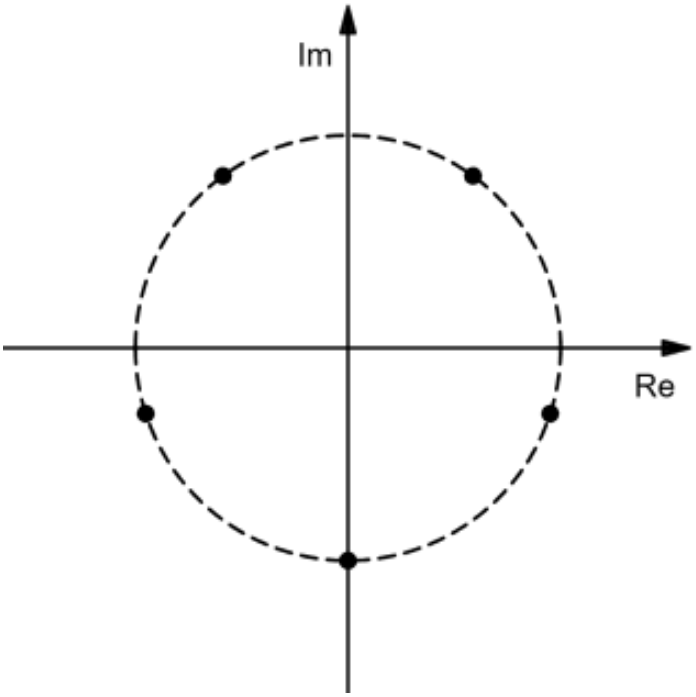
All the roots lie on a unit circle centred at the origin:

1 b)



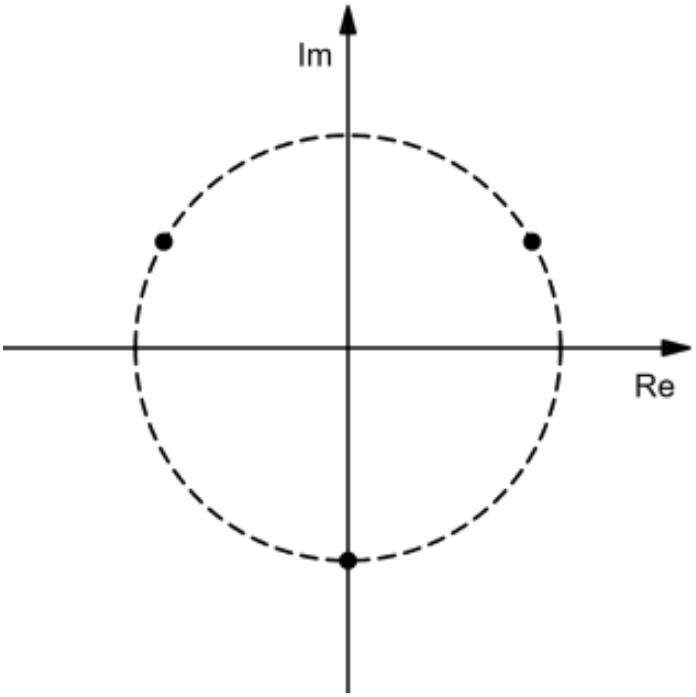
All the roots lie on a unit circle centred at the origin:

1 c)



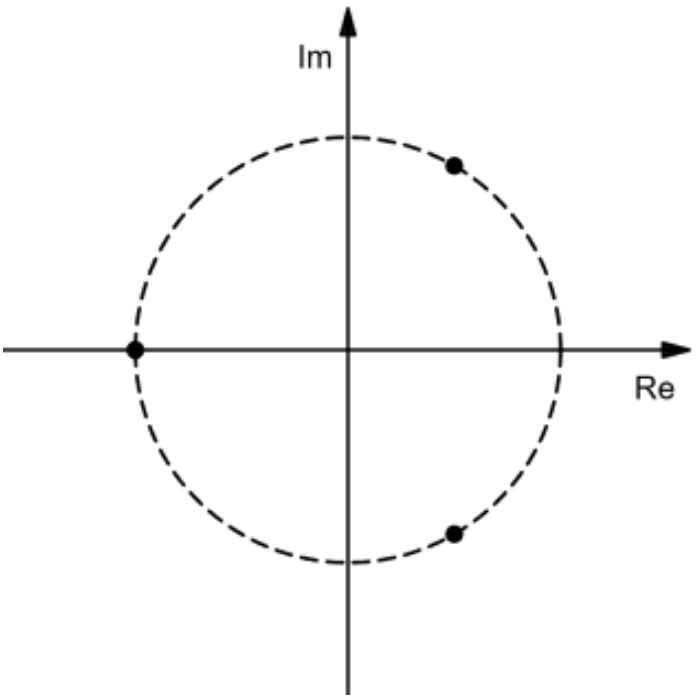
All the roots lie on a unit circle centred at the origin:

1 d)



All the roots lie on a circle of radius 2 centred at the origin:

1 e)



All the roots lie on a circle of radius 2 centred at the origin:

2 a)

$$z = \mathrm{e}^{\pi \mathrm{i}/6}, \quad z = \mathrm{e}^{5\pi \mathrm{i}/6} \quad \text{and} \quad z = \mathrm{e}^{-\pi \mathrm{i}/2}$$
$$\therefore z_0 = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}, \quad z_1 = -\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i} \quad \text{and} \quad z_2 = -\mathrm{i}.$$

2 b)

$$a = 3, \quad b = 1.$$

3 a)

$$z = 4\mathrm{e}^{\pi \mathrm{i}/3}, \quad z = 4\mathrm{e}^{\pi \mathrm{i}} \quad \text{and} \quad z = 4\mathrm{e}^{-\pi \mathrm{i}/3}$$
$$\therefore z_0 = 2 + 2\sqrt{3}\mathrm{i}, \quad z_1 = -4 \quad \text{and} \quad z_2 = 2 - 2\sqrt{3}\mathrm{i}.$$

3 b)

$$a = 3, \quad b = -64.$$

4 a)

$$z_0 = \sqrt{3}\mathrm{e}^{\pi \mathrm{i}/8}, \quad z_1 = \sqrt{3}\mathrm{e}^{5\pi \mathrm{i}/8}, \quad z_2 = \sqrt{3}\mathrm{e}^{9\pi \mathrm{i}/8} \quad \text{and} \quad z_3 = \sqrt{3}\mathrm{e}^{13\pi \mathrm{i}/8}$$
$$\therefore z = \sqrt{3}\mathrm{e}^{\pi \mathrm{i}/8}, \quad z = \sqrt{3}\mathrm{e}^{5\pi \mathrm{i}/8}, \quad z = \sqrt{3}\mathrm{e}^{-7\pi \mathrm{i}/8} \quad \text{and} \quad z = \sqrt{3}\mathrm{e}^{-3\pi \mathrm{i}/8}.$$

4 b)

$$a = 4, \quad b = 9.$$

5 a)

$$z^3 = 8$$

Using modulus-argument form:

$$8 = 8(\cos 0 + \mathrm{i} \sin 0)$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 8 \Rightarrow r = 2$$

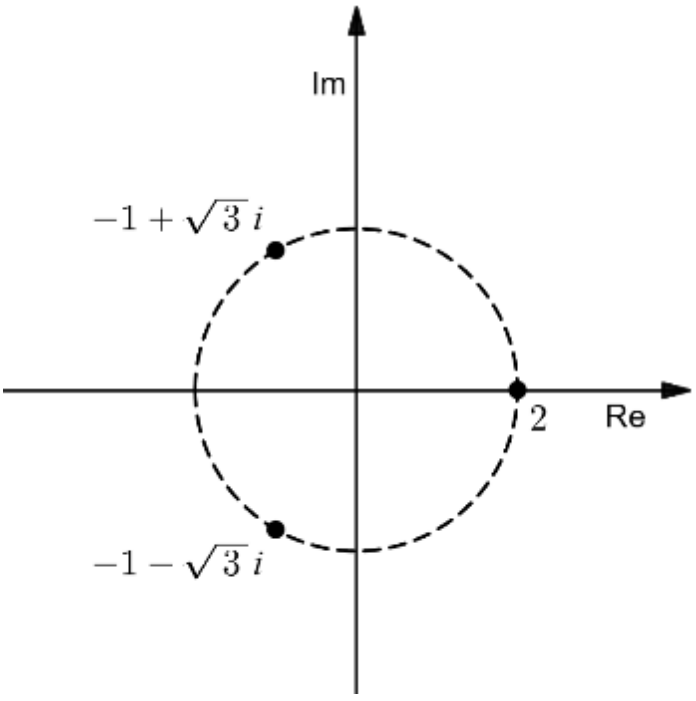
Since $3\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{3}$$

$$\begin{aligned} k = 0: &\quad z_0 = 2\mathrm{e}^{0\mathrm{i}} = 2 \\ k = 1: &\quad z_1 = 2\mathrm{e}^{2\pi\mathrm{i}/3} = -1 + \sqrt{3}\mathrm{i} \\ k = 2: &\quad z_2 = 2\mathrm{e}^{4\pi\mathrm{i}/3} = 2\mathrm{e}^{-2\pi\mathrm{i}/3} = -1 - \sqrt{3}\mathrm{i} \end{aligned}$$

$$\therefore z = 2, \quad z = -1 + \sqrt{3}\mathrm{i} \quad \text{and} \quad z = -1 - \sqrt{3}\mathrm{i}.$$

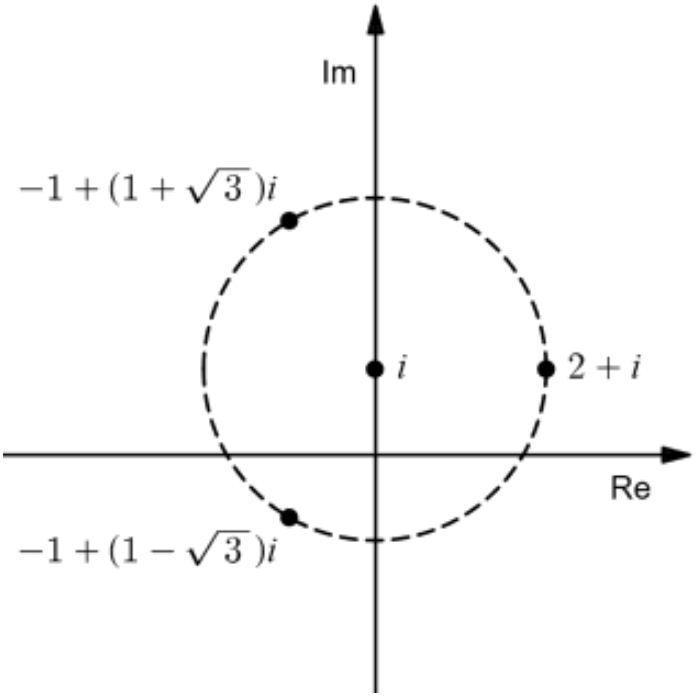
5 b)



5 c)

$$z = 2 + \mathrm{i}, \quad z = \quad \text{and} \quad z = -1 + (1 - \sqrt{3})\mathrm{i}.$$

5 d)



6 a)

$$z^4 = 16\mathrm{i}$$

Using modulus-argument form:

$$16\mathrm{i} = 16(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2})$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$

$$r^4 = 16 \Rightarrow r = 2$$

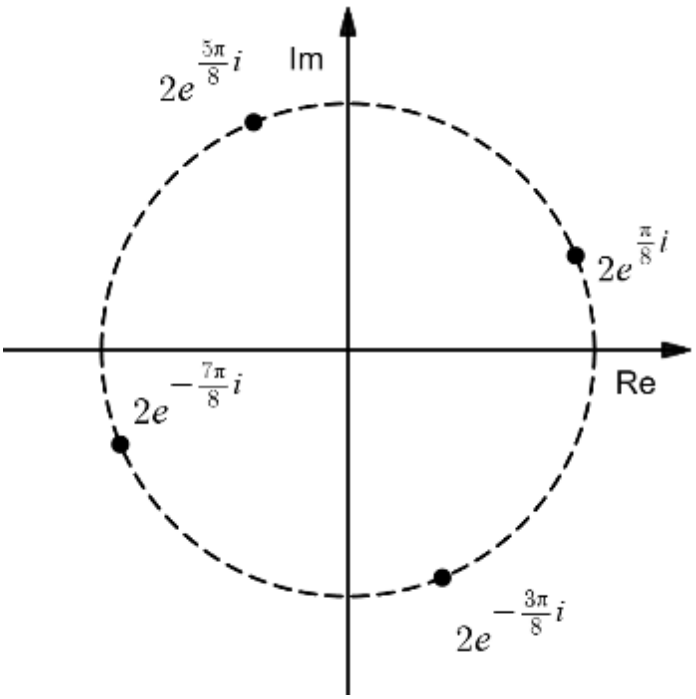
Since $4\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{8} + \frac{2k\pi}{4}$$

$$\begin{aligned} k = 0: &\quad z_0 = 2\mathrm{e}^{\pi\mathrm{i}/8} \\ k = 1: &\quad z_1 = 2\mathrm{e}^{5\pi\mathrm{i}/8} \\ k = 2: &\quad z_2 = 2\mathrm{e}^{9\pi\mathrm{i}/8} = 2\mathrm{e}^{-7\pi\mathrm{i}/8} \\ k = 3: &\quad z_3 = 2\mathrm{e}^{13\pi\mathrm{i}/8} = 2\mathrm{e}^{-3\pi\mathrm{i}/8} \end{aligned}$$

$$\therefore z = 2\mathrm{e}^{\pi\mathrm{i}/8}, \quad z = 2\mathrm{e}^{5\pi\mathrm{i}/8}, \quad z = 2\mathrm{e}^{-7\pi\mathrm{i}/8} \quad \text{and} \quad z = 2\mathrm{e}^{-3\pi\mathrm{i}/8}.$$

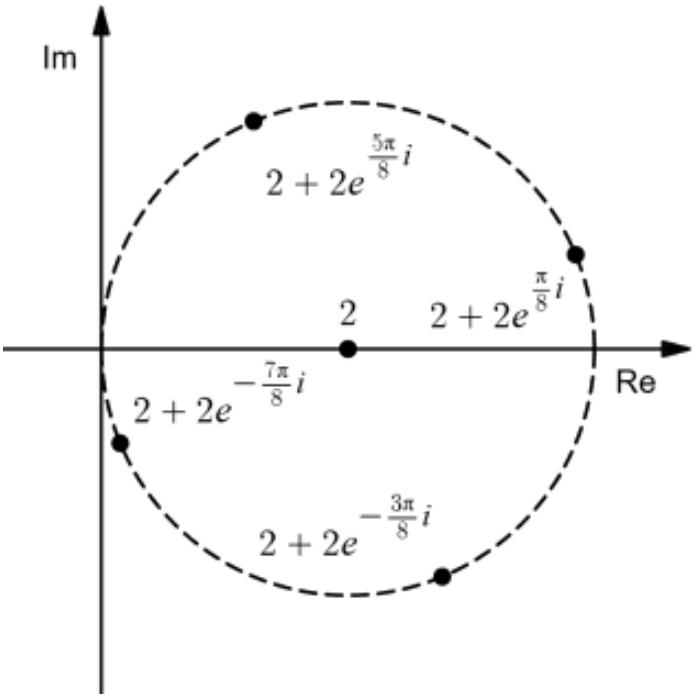
6 b)



6 c)

$$z = 2 + 2\mathrm{e}^{\pi \mathrm{i}/8}, \quad z = 2 + 2\mathrm{e}^{5\pi \mathrm{i}/8}, \quad z = 2 + 2\mathrm{e}^{-7\pi \mathrm{i}/8} \quad \text{and} \quad z = 2 + 2\mathrm{e}^{-3\pi \mathrm{i}/8}.$$

6 d)



7 a)

$$z^6 = -64\mathrm{i}$$

Using modulus-argument form:

$$-64\mathrm{i} = 64(\cos(-\tfrac{\pi}{2}) + \mathrm{i}\sin(-\tfrac{\pi}{2}))$$

$$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)$$

$$r^6 = 64 \Rightarrow r = 2$$

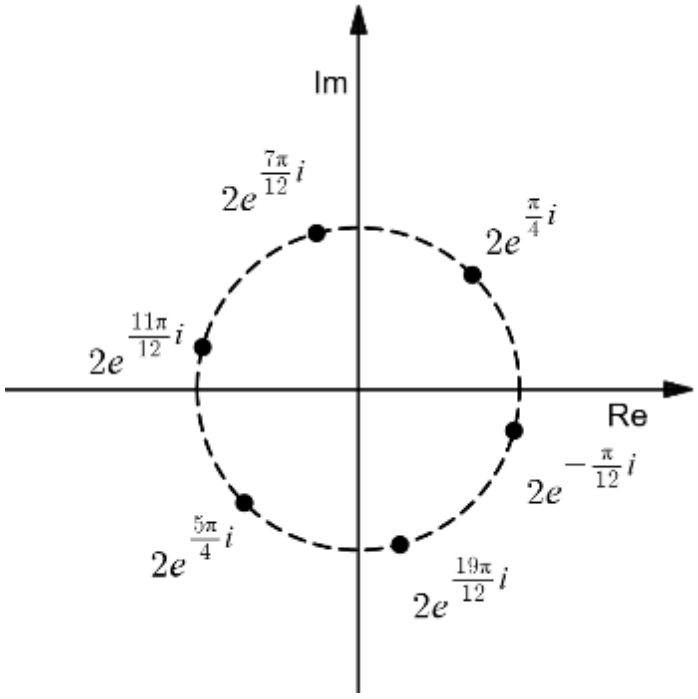
Since $6\theta = -\frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = -\frac{\pi}{12} + \frac{2k\pi}{6}$$

$$\begin{aligned}
k = 0: &\quad z_0 = 2\mathrm{e}^{-\pi\mathrm{i}/12} \\
k = 1: &\quad z_1 = 2\mathrm{e}^{\pi\mathrm{i}/4} \\
k = 2: &\quad z_2 = 2\mathrm{e}^{7\pi\mathrm{i}/12} \\
k = 3: &\quad z_3 = 2\mathrm{e}^{11\pi\mathrm{i}/12} \\
k = 4: &\quad z_4 = 2\mathrm{e}^{5\pi\mathrm{i}/4} = 2\mathrm{e}^{-\pi\mathrm{i}/4} \\
k = 5: &\quad z_5 = 2\mathrm{e}^{19\pi\mathrm{i}/12} = 2\mathrm{e}^{-5\pi\mathrm{i}/12}
\end{aligned}$$

$\therefore z = 2\mathrm{e}^{-\pi\mathrm{i}/12}, \quad z = 2\mathrm{e}^{\pi\mathrm{i}/4}, \quad z = 2\mathrm{e}^{7\pi\mathrm{i}/12}, \quad z = 2\mathrm{e}^{11\pi\mathrm{i}/12}, \quad z = 2\mathrm{e}^{-\pi\mathrm{i}/4} \quad \text{and} \quad z = 2\mathrm{e}^{-5\pi\mathrm{i}/12}.$

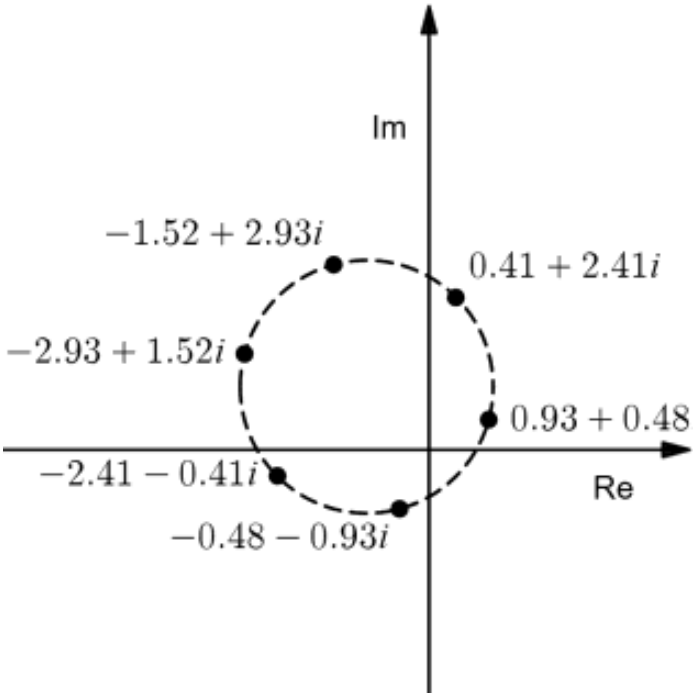
7 b)



7 c)

$$z = 0.93 + 0.48\mathrm{i}, \quad z = 0.41 + 2.41\mathrm{i}, \quad z = -1.52 + 2.93\mathrm{i}, \quad z = -2.93 + 1.52\mathrm{i},$$
$$z = -2.41 - 0.41\mathrm{i} \quad \text{and} \quad z = -0.48 - 0.93\mathrm{i}.$$

7 d)



8 a)

$$|1 + 3\text{i}| = \sqrt{1^2 + 3^2} = \sqrt{10} \quad \text{and} \quad \arg(1 + 3\text{i}) = \arctan(3).$$

To find the angle of the other two vertices, add on $\frac{2\pi}{3}$.

$$\begin{aligned}
\theta_1 &= \arctan(3) + \frac{2\pi}{3} \\
z_1 &= \sqrt{10}\left(\cos\left(\arctan(3) + \frac{2\pi}{3}\right) + \text{i}\sin\left(\arctan(3) + \frac{2\pi}{3}\right)\right) \\
&= -3.10 - 0.634\text{i} \\
\theta_2 &= \arctan(3) + \frac{4\pi}{3} \\
z_2 &= \sqrt{10}\left(\cos\left(\arctan(3) + \frac{4\pi}{3}\right) + \text{i}\sin\left(\arctan(3) + \frac{4\pi}{3}\right)\right) \\
&= 2.10 - 2.37\text{i}
\end{aligned}$$

$$\therefore (1, 3), \quad (-3.10, -0.634) \quad \text{and} \quad (2.10, -2.37).$$

8 b)

$$|3 + 2\text{i}| = \sqrt{9 + 4} = \sqrt{13} \quad \text{and} \quad \arg(3 + 2\text{i}) = \arctan\left(\frac{2}{3}\right).$$

To find the angles of the other vertices, add on $\frac{2\pi}{5}$.

$$\begin{aligned}
\theta_1 &= \arctan\left(\frac{2}{3}\right) + \frac{2\pi}{5} \\
z_1 &= \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right) + \frac{2\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right) + \frac{2\pi}{5}\right)\right)
\end{aligned}$$

$$
\begin{aligned}
&= -0.975 + 3.47\text{i} \\
\theta_2 &= \arctan\left(\frac{2}{3}\right) + \frac{4\pi}{5} \\
z_2 &= \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right) + \frac{4\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right) + \frac{4\pi}{5}\right)\right) \\
&= -3.60 + 0.145\text{i} \\
\theta_3 &= \arctan\left(\frac{2}{3}\right) + \frac{6\pi}{5} \\
z_3 &= \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right) + \frac{6\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right) + \frac{6\pi}{5}\right)\right) \\
&= -1.25 - 3.38\text{i} \\
\theta_4 &= \arctan\left(\frac{2}{3}\right) + \frac{8\pi}{5} \\
z_4 &= \sqrt{13}\left(\cos\left(\arctan\left(\frac{2}{3}\right) + \frac{8\pi}{5}\right) + \text{i}\sin\left(\arctan\left(\frac{2}{3}\right) + \frac{8\pi}{5}\right)\right) \\
&= 2.83 - 2.24\text{i}
\end{aligned}
$$

$$\therefore (3, 2), \quad (-0.975, 3.47), \quad (-3.60, 0.145), \quad (-1.25, -3.38) \quad \text{and} \quad (2.83, -2.24).$$

8 c)

$$|2 + 2\text{i}| = \sqrt{4 + 4} = 2\sqrt{2} \quad \text{and} \quad \arg(2 + 2\text{i}) = \arctan(1).$$

To find the angle of the other vertices, add on $\frac{2\pi}{6}$.

$$
\begin{aligned}
\theta_1 &= \arctan(1) + \frac{2\pi}{6} \\
z_1 &= 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{2\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{2\pi}{6}\right)\right) \\
&= -0.732 + 2.73\text{i} \\
\theta_2 &= \arctan(1) + \frac{4\pi}{6} \\
z_2 &= 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{4\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{4\pi}{6}\right)\right) \\
&= -2.73 + 0.732\text{i} \\
\theta_3 &= \arctan(1) + \frac{6\pi}{6} \\
z_3 &= 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{6\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{6\pi}{6}\right)\right) \\
&= -2 - 2\text{i} \\
\theta_4 &= \arctan(1) + \frac{8\pi}{6} \\
z_4 &= 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{8\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{8\pi}{6}\right)\right) \\
&= 0.732 - 2.73\text{i} \\
\theta_5 &= \arctan(1) + \frac{10\pi}{6} \\
z_5 &= 2\sqrt{2}\left(\cos\left(\arctan(1) + \frac{10\pi}{6}\right) + \text{i}\sin\left(\arctan(1) + \frac{10\pi}{6}\right)\right) \\
&= 2.73 - 0.732\text{i}
\end{aligned}
$$

$\therefore (2,2), \quad (-0.732, 2.73), \quad (-2.73, 0.732), \quad (-2,-2), \quad (0.732, -2.73) \quad \text{and} \quad (2.73, -0.732).$

9 a)

$$|2+2\mathrm{i}| = \sqrt{4+4} = 2\sqrt{2}.$$

Angle between the vertices: $\frac{2\pi}{3}$.

$$\text{Area} = 3 \times \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2} \times \sin\left(\frac{2\pi}{3}\right) = 6\sqrt{3} \text{ units}^2.$$

9 b)

$$|5+\mathrm{i}| = \sqrt{25+1} = \sqrt{26}.$$

Angle between the vertices: $\frac{2\pi}{4}$.

$$\text{Area} = 4 \times \frac{1}{2} \times \sqrt{26} \times \sqrt{26} \times \sin\left(\frac{2\pi}{4}\right) = 52 \text{ units}^2.$$

9 c)

$$|3+4\mathrm{i}| = \sqrt{9+16} = 5.$$

Angle between the vertices: $\frac{2\pi}{6}$.

$$\text{Area} = 6 \times \frac{1}{2} \times 5 \times 5 \times \sin\left(\frac{2\pi}{6}\right) = \frac{75}{2}\sqrt{3} \text{ units}^2.$$

10)

Translate $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ so the vertex at $(2,2)$ becomes a vertex at $(2,-1)$.

$$|2-\mathrm{i}| = \sqrt{4+1} = \sqrt{5} \quad \text{and} \quad \arg(2-\mathrm{i}) = -\arctan\left(\frac{1}{2}\right).$$

To find the angles of the other vertices, add on $\frac{2\pi}{6}$.

$$\begin{aligned}
\theta_1 &= -\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6} \\
z_1 &= \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{2\pi}{6}\right)\right) \\
&= 1.87 + 1.23\mathrm{i} \\
\theta_2 &= -\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6} \\
z_2 &= \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{4\pi}{6}\right)\right) = -0.13 + 2.23\mathrm{i} \\
\theta_3 &= -\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6} \\
z_3 &= \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{6\pi}{6}\right)\right) = -2 + \mathrm{i}
\end{aligned}$$

$$\theta_4 = -\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}$$
$$z_4 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{8\pi}{6}\right)\right) = -1.87 - 1.23\mathrm{i}$$
$$\theta_5 = -\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}$$
$$z_5 = \sqrt{5}\left(\cos\left(-\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}\right) + \mathrm{i}\sin\left(-\arctan\left(\frac{1}{2}\right) + \frac{10\pi}{6}\right)\right) = 0.13 - 2.23\mathrm{i}$$

Translate the points back:

$$\therefore (2,2), \quad (1.87, 4.23), \quad (-0.13, 5.23), \quad (-2,4), \quad (-1.87, 1.77) \quad \text{and} \quad (0.13, 0.77).$$

11)

Translate $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ so the vertex at $(2,-1)$ becomes a vertex at $(1,-5)$.

$$|1 - 5\mathrm{i}| = \sqrt{1+25} = \sqrt{26} \quad \text{and} \quad \arg(1 - 5\mathrm{i}) = -\arctan 5.$$

To find the angles of the other vertices, add on $\frac{2\pi}{3}$.

$$\theta_1 = -\arctan 5 + \frac{2\pi}{3}$$
$$z_1 = \sqrt{26}\left(\cos\left(-\arctan 5 + \frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\arctan 5 + \frac{2\pi}{3}\right)\right) = 3.83 + 3.37\mathrm{i}$$
$$\theta_2 = -\arctan 5 + \frac{4\pi}{3}$$
$$z_2 = \sqrt{26}\left(\cos\left(-\arctan 5 + \frac{4\pi}{3}\right) + \mathrm{i}\sin\left(-\arctan 5 + \frac{4\pi}{3}\right)\right) = -4.83 + 1.63\mathrm{i}$$

Translate the points back:

$$\therefore (2,-1), \quad (4.83, 7.37) \quad \text{and} \quad (-3.83, 5.63).$$

12)

$$3\sqrt{3} = 3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right) \quad \text{so} \quad x = 2.$$

Therefore $(2,0)$ is a vertex of the triangle.

$$\theta_1 = \frac{2\pi}{3}, \qquad z_1 = 2(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}) = -1 + \sqrt{3}\mathrm{i}$$
$$\theta_2 = \frac{4\pi}{3}, \qquad z_2 = 2(\cos\tfrac{4\pi}{3} + \mathrm{i}\sin\tfrac{4\pi}{3}) = -1 - \sqrt{3}\mathrm{i}$$

$$\therefore (2,0), \quad (-1, \sqrt{3}) \quad \text{and} \quad (-1, -\sqrt{3}).$$

13)

$$12\sqrt{3} = 3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right) \quad \text{so} \quad x = 4.$$

Therefore $(1, 5)$ is a vertex of the triangle. Translate $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ so the vertex at $(1, 5)$ becomes a vertex at $(0, 4)$.

$$|4\text{i}| = \sqrt{0 + 16} = 4 \quad \text{and} \quad \arg(4\text{i}) = \frac{\pi}{2}.$$

To find the angles of the other vertices, add on $\frac{2\pi}{3}$.

$$\theta_1 = \frac{\pi}{2} + \frac{2\pi}{3}, \qquad z_1 = 4(\cos\tfrac{7\pi}{6} + \text{i}\sin\tfrac{7\pi}{6}) = -2\sqrt{3} - 2\text{i}$$
$$\theta_2 = \frac{\pi}{2} + \frac{4\pi}{3}, \qquad z_2 = 4(\cos\tfrac{11\pi}{6} + \text{i}\sin\tfrac{11\pi}{6}) = 2\sqrt{3} - 2\text{i}$$

Translate the points back:

$$\therefore (1, 5), \quad (1 - 2\sqrt{3}, -1) \quad \text{and} \quad (1 + 2\sqrt{3}, -1).$$

14 a)

$$|3 + 4\text{i}| = \sqrt{9 + 16} = 5 \quad \text{and} \quad \arg(3 + 4\text{i}) = \arctan\frac{4}{3}.$$

To find the angle of the other vertices, add on $\frac{2\pi}{3}$.

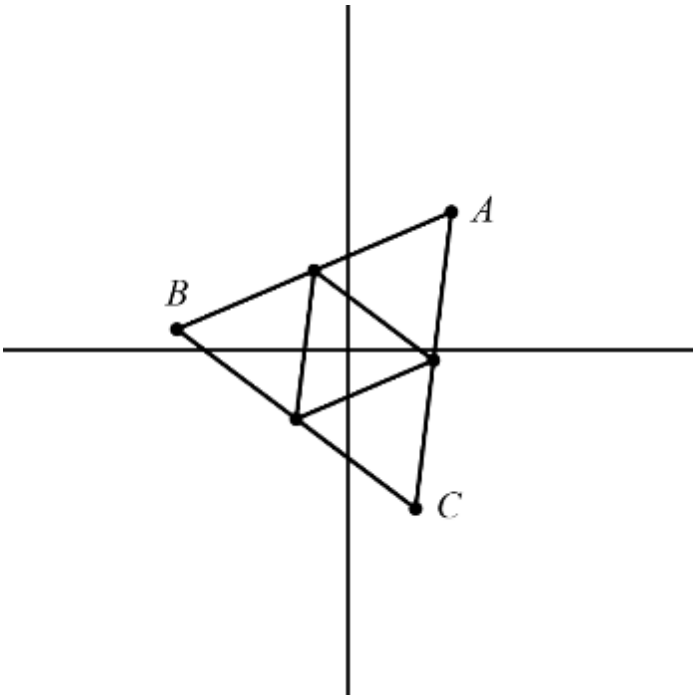
$$\theta_1 = \arctan\frac{4}{3} + \frac{2\pi}{3}$$
$$z_1 = 5\left(\cos\left(\arctan\frac{4}{3} + \frac{2\pi}{3}\right) + \text{i}\sin\left(\arctan\frac{4}{3} + \frac{2\pi}{3}\right)\right) = -4.96 + 0.60\text{i}$$
$$\theta_2 = \arctan\frac{4}{3} + \frac{4\pi}{3}$$
$$z_2 = 5\left(\cos\left(\arctan\frac{4}{3} + \frac{4\pi}{3}\right) + \text{i}\sin\left(\arctan\frac{4}{3} + \frac{4\pi}{3}\right)\right) = 1.96 - 4.60\text{i}$$

$$\therefore B(-4.96, 0.60) \quad \text{and} \quad C(1.96, -4.60).$$

14 b)

$$3 \times \frac{1}{2} \times 5^2 \times \sin\left(\frac{2\pi}{3}\right) = \frac{75}{4}\sqrt{3} \text{ units}^2.$$

14 c)



$$\frac{1}{4} \times 3 \times \frac{1}{2} \times 5^2 \times \sin\left(\frac{2\pi}{3}\right) = \frac{75}{16}\sqrt{3} \text{ units}^2.$$

15)

$$3 \times \frac{1}{2} \times x^2 \times \sin\left(\frac{2\pi}{3}\right) = 27\sqrt{3} \Rightarrow x = 6.$$

So the roots are

$$z_1 = 6\mathrm{e}^{\pi \mathrm{i}/3} = 3 + 3\sqrt{3}\mathrm{i}, \quad z_2 = 6\mathrm{e}^{\pi \mathrm{i}} = -6 \quad \text{and} \quad z_3 = 6\mathrm{e}^{-\pi \mathrm{i}/3} = 3 + 3\sqrt{3}\mathrm{i}.$$

So

$$\begin{aligned} f(z) &= (z+6)(z-(3+3\sqrt{3}\mathrm{i}))(z-(3-3\sqrt{3}\mathrm{i})) \\ &= (z+6)(z^2-6z+36) \\ &= z^3 + 216. \end{aligned}$$

$$\therefore a = 216.$$

16)

$$3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right) = 3\sqrt{3} \Rightarrow x = 2.$$

Since $z_1 = 3 + \mathrm{i}$, translate $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ so the vertex $(3, 1)$ becomes $(2, 0)$.

$$\begin{aligned} \theta_1 &= 0 + \frac{2\pi}{3}, & z_1 &= 2(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}) = -1 + \sqrt{3}\mathrm{i} \\ \theta_2 &= 0 + \frac{4\pi}{3}, & z_2 &= 2(\cos\tfrac{4\pi}{3} + \mathrm{i}\sin\tfrac{4\pi}{3}) = -1 - \sqrt{3}\mathrm{i} \end{aligned}$$

Translate points back:

$$\therefore (3, 1), \quad (0, 1+\sqrt{3}) \quad \text{and} \quad (0, 1-\sqrt{3}).$$

$$z_1 = 3 + \mathrm{i}, \quad z_2 = (1+\sqrt{3})\mathrm{i} \quad \text{and} \quad z_3 = (1-\sqrt{3})\mathrm{i}.$$

$$\begin{aligned} f(z) &= (z-(3+\mathrm{i}))(z-((1+\sqrt{3})\mathrm{i}))(z-((1-\sqrt{3})\mathrm{i})) \\ &= (z-(3+\mathrm{i}))(z^2-2\mathrm{i}z+2) \\ &= z^3-(3+3\mathrm{i})z^2+6\mathrm{i}z-(6+2\mathrm{i}). \end{aligned}$$

$$\therefore a = -3-3\mathrm{i}, \quad b = 6\mathrm{i} \quad \text{and} \quad c = -6-2\mathrm{i}.$$

17)

$$3 \times \frac{1}{2}x^2 \sin\left(\frac{2\pi}{3}\right) = \frac{27}{4}\sqrt{3} \Rightarrow x = 3.$$

Since $z_1 = 6 + 0\mathrm{i}$, translate $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ so the vertex $(6, 0)$ becomes $(3, 0)$.

$$\theta_1 = 0 + \frac{2\pi}{3}, \qquad z_1 = 3(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}) = -\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}$$

$$\theta_2 = 0 + \frac{4\pi}{3}, \qquad z_2 = 3(\cos\tfrac{4\pi}{3} + \mathrm{i}\sin\tfrac{4\pi}{3}) = -\frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$$

Translate points back:

$$z_0 = 6 + 0\mathrm{i}, \quad z_1 = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i} \quad \text{and} \quad z_2 = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}.$$

$$\begin{aligned} f(z) &= \left(z-(\tfrac{3}{2}+\tfrac{3\sqrt{3}}{2}\mathrm{i})\right)\left(z-(\tfrac{3}{2}-\tfrac{3\sqrt{3}}{2}\mathrm{i})\right)(z-6) \\ &= (z^2-3z+9)(z-6) \\ &= z^3-9z^2+27z-54. \end{aligned}$$

$$\therefore a = -9, \quad b = 27 \quad \text{and} \quad c = -54.$$

18 a)

Solve $w^3 = 27\mathrm{i}$ where $w = z - 1$. Using modulus-argument form:

$$27\mathrm{i} = 27(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2})$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 27 \Rightarrow r = 3$$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$$

$$k = 0: \quad z_0 = 3(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}) = \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$$

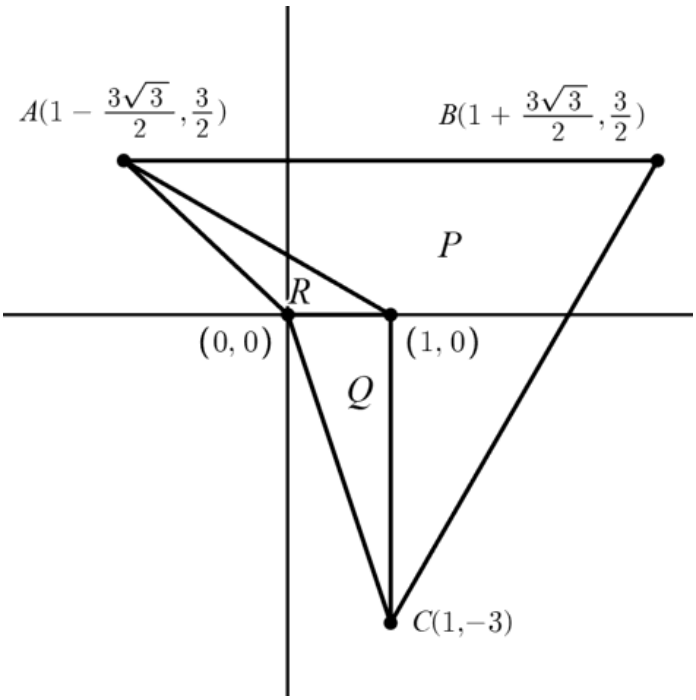
$$k = 1: \quad z_1 = 3(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6}) = -\frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$$

$$k = 2: \quad z_2 = 3(\cos(-\tfrac{\pi}{2}) + \mathrm{i}\sin(-\tfrac{\pi}{2})) = -3\mathrm{i}$$

So $w = \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$, $w = \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}$ and $w = -3\mathrm{i}$, and so $z = w + 1$.

$$\therefore w = 1 + \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}, \quad w = 1 - \frac{3\sqrt{3}}{2} + \frac{3}{2}\mathrm{i} \quad \text{and} \quad w = 1 - 3\mathrm{i}.$$

18 b)



Centre is at $(1,0)$ with vertices

$$\left(1+\frac{3\sqrt{3}}{2},\frac{3}{2}\right), \quad \left(1-\frac{3\sqrt{3}}{2},\frac{3}{2}\right) \quad \text{and} \quad (1,-3).$$

$$\text{Area } P = 2\times\frac{1}{2}\times 3^2\times\sin\frac{2\pi}{3} = \frac{9\sqrt{3}}{2}$$
$$\text{Area } Q = \frac{1}{2}\times 1\times 3 = \frac{3}{2}$$
$$\text{Area } R = \frac{1}{2}\times 1\times\frac{3}{2} = \frac{3}{4}$$

Total area:

$$\frac{9\sqrt{3}}{2}+\frac{9}{4} \quad \text{as required.}$$

19 a)

$$z^3 = -5+\sqrt{75}\mathrm{i}$$

Using modulus-argument form:

$$-5+\sqrt{75}\mathrm{i} = 10(\cos\tfrac{2\pi}{3}+\mathrm{i}\sin\tfrac{2\pi}{3})$$
$$z^3 = r^3(\cos\theta+\mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta+\mathrm{i}\sin 3\theta)$$
$$r^3 = 10 \Rightarrow r = 10^{1/3}$$

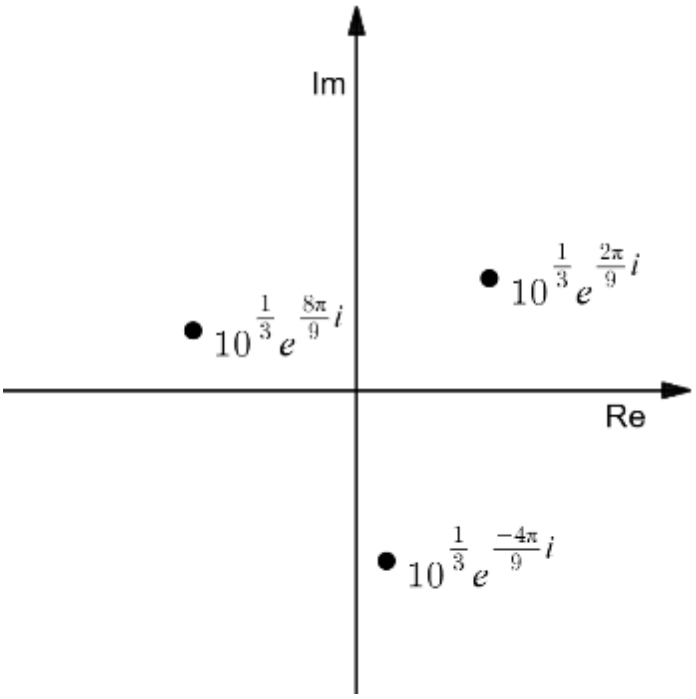
Since $3\theta = \frac{2\pi}{3}+2k\pi$ for integer k:

$$\theta = \frac{2\pi}{9}+\frac{2k\pi}{3}$$

$$k = 0: \quad z_0 = 10^{1/3}\mathrm{e}^{2\pi\mathrm{i}/9}$$
$$k = 1: \quad z_1 = 10^{1/3}\mathrm{e}^{8\pi\mathrm{i}/9}$$
$$k = 2: \quad z_2 = 10^{1/3}\mathrm{e}^{14\pi\mathrm{i}/9} = 10^{1/3}\mathrm{e}^{-4\pi\mathrm{i}/9}$$

$$\therefore z = 10^{1/3}\mathrm{e}^{2\pi\mathrm{i}/9}, \quad z = 10^{1/3}\mathrm{e}^{8\pi\mathrm{i}/9} \quad \text{and} \quad z = 10^{1/3}\mathrm{e}^{-4\pi\mathrm{i}/9}.$$

19 b)



19 c)

$$\frac{2\pi}{9}, \quad \frac{8\pi}{9} \quad \text{and} \quad -\frac{4\pi}{9}.$$

20)

$$\frac{\sqrt{3}}{2}(1+\mathrm{i}) = \sqrt{\frac{3}{2}}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}) \quad \text{or} \quad \frac{\sqrt{6}}{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}).$$
$$\therefore \text{ the least value } \arg(z) = \frac{\pi/4}{5} = \frac{\pi}{20}.$$

21)

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} = \cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}.$$
$$\therefore \text{ the least value } \arg(z) = \frac{2\pi/3}{6} = \frac{\pi}{9}.$$

22 a)

$$\mathrm{i} = \cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}.$$

Since $6\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{6}.$$

$$k = 0: \quad z_0 \text{ has argument } \frac{\pi}{12}$$
$$k = 1: \quad z_1 \text{ has argument } \frac{5\pi}{12}$$

Mid-angle of z_0 and z_1 is

$$\frac{1}{2}\left(\frac{\pi}{12} + \frac{5\pi}{12}\right) = \frac{\pi}{4}.$$

Since $w = \frac{1}{2}(z_1 + z_2)$, $|z_1| = |z_2|$ and their arguments are symmetric about their midpoint, the sum lies in the direction of the angle bisector, so that

$$\arg\omega = \frac{\pi}{4}.$$

22 b)

For ω^n to be real:

$$\frac{n\pi}{4} = k\pi \text{ for some integer } k \Rightarrow n = 4k, \text{ so let } k = 1.$$
$$\therefore n = 4.$$

23 a)

$$\sqrt{3} + \mathrm{i} = 2(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}).$$

Since $5\theta = \frac{\pi}{6} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{30} + \frac{2k\pi}{5}.$$

$$k = 0: \quad z_0 \text{ has argument } \frac{\pi}{30}$$
$$k = 1: \quad z_1 \text{ has argument } \frac{13\pi}{30}$$

Mid-angle of z_0 and z_1 is

$$\frac{1}{2}\left(\frac{\pi}{30} + \frac{13\pi}{30}\right) = \frac{7\pi}{30}.$$

Since $w = \frac{1}{2}(z_1 + z_2)$, $|z_1| = |z_2|$ and their arguments are symmetric about their midpoint, the sum lies in the direction of the angle bisector, so that

$$\arg\omega = \frac{7\pi}{30}.$$

23 b)

For ω^n to be real:

$$\frac{7n\pi}{30} = k\pi \text{ for some integer } k \Rightarrow n = \frac{30k}{7}, \text{ so let } k = 7.$$
$$\therefore n = 30.$$

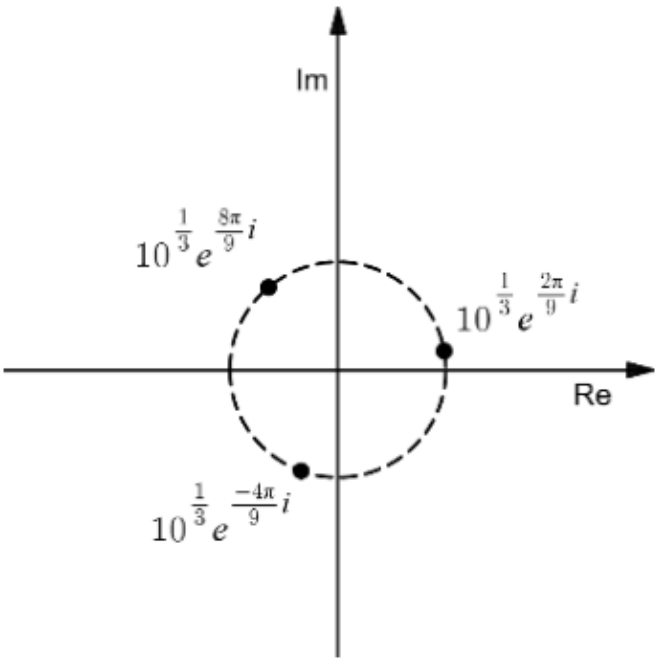
24 a)

$$z^3 = \sqrt{6} + \sqrt{2}\mathrm{i} = 2\sqrt{2}\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right).$$

$$r^3 = 2\sqrt{2} \Rightarrow r = \sqrt{2}, \qquad 3\theta = \frac{\pi}{6} + 2k\pi \Rightarrow \theta = \frac{\pi}{18} + \frac{2k\pi}{3}.$$

$$k = 0: \ z = \sqrt{2}\,\mathrm{cis}\,\frac{\pi}{18}, \quad k = 1: \ z = \sqrt{2}\,\mathrm{cis}\,\frac{13\pi}{18}, \quad k = 2: \ z = \sqrt{2}\,\mathrm{cis}\left(-\frac{11\pi}{18}\right).$$

24 b)



24 c)

The determinant of the matrix is

$$\det M = 24 - 4 = 20.$$

Area of triangle formed by roots:

$$3 \times \frac{1}{2} \times (\sqrt{2})^2 \sin\frac{2\pi}{3} = \frac{3}{2}\sqrt{3}.$$

$$\therefore \ \text{transformed area} = 20 \times \frac{3\sqrt{3}}{2} = 30\sqrt{3}.$$

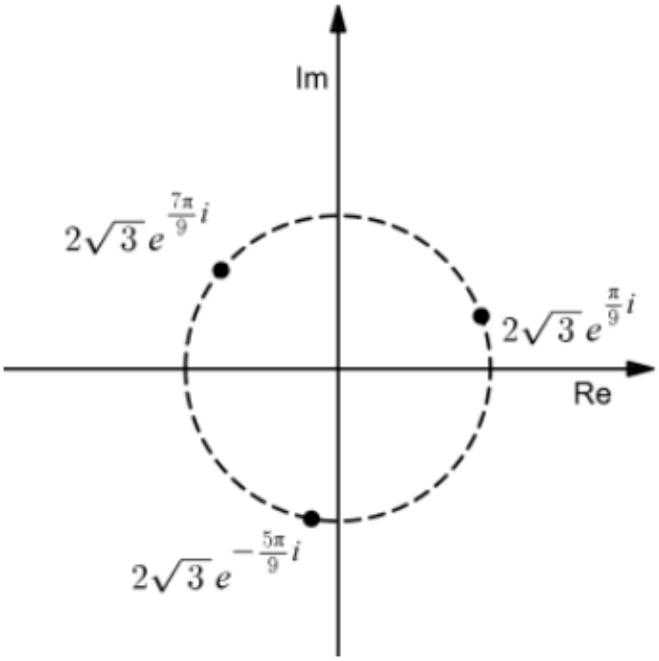
25 a)

$$z^3 = 12\sqrt{3} + 36\mathrm{i} = 24\sqrt{3}\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right).$$

$$r^3 = 24\sqrt{3} \Rightarrow r = 2\sqrt{3}, \qquad 3\theta = \frac{\pi}{3} + 2k\pi \Rightarrow \theta = \frac{\pi}{9} + \frac{2k\pi}{3}.$$

$$k = 0: \ z = 2\sqrt{3}\,\mathrm{cis}\,\frac{\pi}{9}, \quad k = 1: \ z = 2\sqrt{3}\,\mathrm{cis}\,\frac{7\pi}{9}, \quad k = 2: \ z = 2\sqrt{3}\,\mathrm{cis}\left(-\frac{5\pi}{9}\right).$$

25 b)



25 c)

The determinant is

$$\begin{vmatrix} 2 & -2 \\ 2 & 2 \end{vmatrix} = 8.$$

The area of the original triangle is

$$\frac{1}{2}(3\sqrt{3})(6) = 9\sqrt{3}.$$

$$\therefore \text{ transformed area} = 8(9\sqrt{3}) = 72\sqrt{3}.$$

26 a)

$$z^6 = 27 \Rightarrow r = \sqrt{3}, \qquad 6\theta = 2k\pi.$$

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

The roots printed in the source are

$$\sqrt{3}, \quad \frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}, \quad -\frac{\sqrt{3}}{2} + \frac{3}{2}\mathrm{i}, \quad -\sqrt{3}, \quad -\frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}, \quad \frac{\sqrt{3}}{2} - \frac{3}{2}\mathrm{i}.$$

26 b)

$$\text{Area} = \frac{9\sqrt{3}}{2}.$$

26 c)

The transformation maps

$$w_0 = \sqrt{3} \quad \text{to} \quad 3 + \sqrt{3}\mathrm{i}.$$

$$|w_0| = 2\sqrt{3}, \qquad \arg(w_0) = \frac{\pi}{6}.$$

$$w_0^6 = (2\sqrt{3})^6 \left(\cos\pi + \mathrm{i}\sin\pi\right) = -1728.$$

For $w^6 + k = 0$,

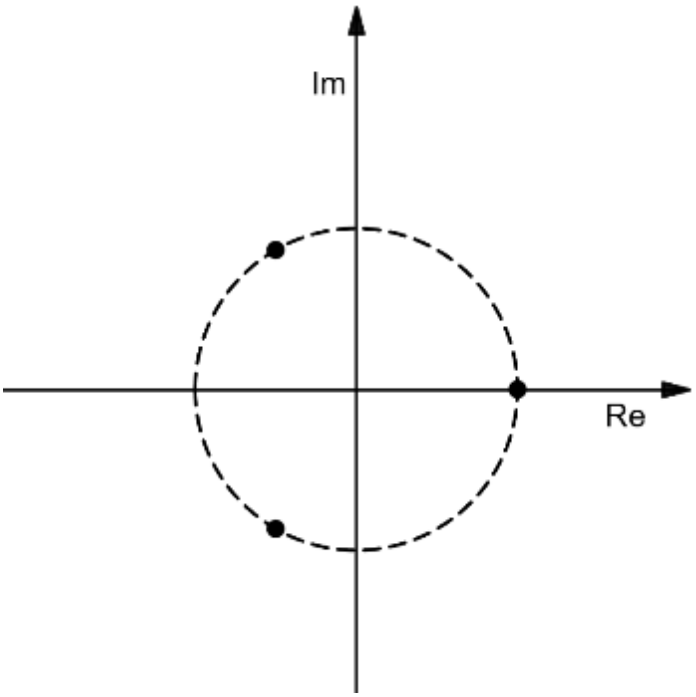
$$k = 1728.$$

26 d)

The determinant is 4, so the area is multiplied by 4:

$$4 \times \frac{9\sqrt{3}}{2} = 18\sqrt{3}.$$

27 a)



27 b)

The three cube roots of unity are equally spaced points on the unit circle forming a regular triangle centred at the origin. The centroid of a regular triangle is at the origin, so the vector sum of the three position vectors is zero.

27 c)

$$z^3 = 1 + \mathrm{i} = \sqrt{2}\left(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}\right).$$

$$r^3 = \sqrt{2} \Rightarrow r = (\sqrt{2})^{1/3} = 2^{1/6}.$$

Since $3\theta = \frac{\pi}{4} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{3}.$$

$$\begin{aligned}
k = 0: &\quad z_0 = 2^{1/6}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right)\\
k = 1: &\quad z_1 = 2^{1/6}\left(\cos\frac{9\pi}{12} + \mathrm{i}\sin\frac{9\pi}{12}\right)\\
k = 2: &\quad z_2 = 2^{1/6}\left(\cos\frac{17\pi}{12} + \mathrm{i}\sin\frac{17\pi}{12}\right) = 2^{1/6}\left(\cos\left(-\frac{7\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{7\pi}{12}\right)\right).
\end{aligned}$$

$$\therefore z = 2^{1/6}\operatorname{cis}\frac{\pi}{12}, \quad z = 2^{1/6}\operatorname{cis}\frac{3\pi}{4}, \quad z = 2^{1/6}\operatorname{cis}\left(-\frac{7\pi}{12}\right).$$

27 d)

Consider the sum of roots which is zero, so the imaginary part is also zero:

$$2^{1/6}\sin\frac{\pi}{12} + 2^{1/6}\sin\frac{9\pi}{12} + 2^{1/6}\sin\left(-\frac{7\pi}{12}\right) = 0.$$

$$\sin\frac{\pi}{12} + \sin\frac{9\pi}{12} + \sin\left(-\frac{7\pi}{12}\right) = 0.$$

$$\sin 15^\circ + \sin 135^\circ + \sin(-105^\circ) = 0.$$

But since sine is an odd function:

$$\sin 15^\circ + \sin 135^\circ - \sin 105^\circ = 0.$$

$$\therefore \sin 15^\circ + \sin 135^\circ = \sin 105^\circ \text{ as req.}$$

28 a)

$$z^4 = 2\sqrt{3} + 2\mathrm{i} = 4\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right).$$

$$r^4 = 4 \Rightarrow r = \sqrt{2}.$$

Since $4\theta = \frac{\pi}{6} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{24} + \frac{2k\pi}{4}.$$

$$\begin{aligned}
k = 0: &\quad z_0 = \sqrt{2}\operatorname{cis}\frac{\pi}{24}\\
k = 1: &\quad z_1 = \sqrt{2}\operatorname{cis}\frac{13\pi}{24}\\
k = 2: &\quad z_2 = \sqrt{2}\operatorname{cis}\frac{25\pi}{24} = \sqrt{2}\operatorname{cis}\left(-\frac{23\pi}{24}\right)\\
k = 3: &\quad z_3 = \sqrt{2}\operatorname{cis}\frac{37\pi}{24} = \sqrt{2}\operatorname{cis}\left(-\frac{11\pi}{24}\right).
\end{aligned}$$

$$\therefore z = \sqrt{2}\operatorname{cis}\frac{\pi}{24},\ \sqrt{2}\operatorname{cis}\frac{13\pi}{24},\ \sqrt{2}\operatorname{cis}\left(-\frac{23\pi}{24}\right),\ \sqrt{2}\operatorname{cis}\left(-\frac{11\pi}{24}\right).$$

28 b)

Consider the sum of roots which is zero, so the real part is also zero:

$$\sqrt{2}\cos\frac{\pi}{24}+\sqrt{2}\cos\frac{13\pi}{24}+\sqrt{2}\cos\left(-\frac{23\pi}{24}\right)+\sqrt{2}\cos\left(-\frac{11\pi}{24}\right)=0.$$

$$\cos\frac{\pi}{24}+\cos\frac{13\pi}{24}+\cos\left(-\frac{23\pi}{24}\right)+\cos\left(-\frac{11\pi}{24}\right)=0.$$

$$\cos 7.5^\circ+\cos 97.5^\circ+\cos(-172.5^\circ)+\cos(-82.5^\circ)=0.$$

But since cosine is an even function:

$$\cos 7.5^\circ+\cos 97.5^\circ+\cos 172.5^\circ+\cos 82.5^\circ=0.$$

$$\therefore \cos 7.5^\circ+\cos 82.5^\circ+\cos 97.5^\circ+\cos 172.5^\circ=0 \text{ as req.}$$

29 a)

$$z^5=3^5 \quad \text{so} \quad z^5=243 \quad \text{or} \quad z^5-243=0.$$

29 b)

$$z^5=243.$$

Using modulus-argument form:

$$243=243(\cos 0+\mathrm{i}\sin 0).$$

$$z^5=r^5(\cos\theta+\mathrm{i}\sin\theta)^5=r^5(\cos 5\theta+\mathrm{i}\sin 5\theta).$$

$$r=3.$$

Since $5\theta=0+2k\pi$ for integer k:

$$\theta=0+\frac{2k\pi}{5}.$$

$$\begin{aligned}
k=0: &\quad z_0=3(\cos 0+\mathrm{i}\sin 0)=3\\
k=1: &\quad z_1=3\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)\\
k=2: &\quad z_2=3\left(\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}\right)\\
k=3: &\quad z_3=3\left(\cos\frac{6\pi}{5}+\mathrm{i}\sin\frac{6\pi}{5}\right)=3\left(\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right)\\
k=4: &\quad z_4=3\left(\cos\frac{8\pi}{5}+\mathrm{i}\sin\frac{8\pi}{5}\right)=3\left(\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right).
\end{aligned}$$

$$\therefore z=3, \quad z=3\,\mathrm{cis}\,\frac{2\pi}{5}, \quad z=3\,\mathrm{cis}\,\frac{4\pi}{5}, \quad z=3\,\mathrm{cis}\left(-\frac{4\pi}{5}\right), \quad z=3\,\mathrm{cis}\left(-\frac{2\pi}{5}\right).$$

29 c)

$$z_B = 3\,\mathrm{cis}\,\frac{2\pi}{5} \quad \text{and} \quad z_E = 3\,\mathrm{cis}\left(-\frac{2\pi}{5}\right).$$

Summing z_B and z_E:

$$\begin{aligned}
&3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) + 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right) \\
&= 3\left[\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\left(-\frac{2\pi}{5}\right) - \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right] \\
&= 3\left[\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right] = 3 \times 2\cos\frac{2\pi}{5}.
\end{aligned}$$

Multiplying z_B and z_E:

$$\begin{aligned}
&3\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) 3\left(\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)\right) \\
&= 9\left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right)\left(\cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right) \\
&= 9\left(\cos^2\frac{2\pi}{5} + \sin^2\frac{2\pi}{5}\right) = 9.
\end{aligned}$$

So that

$$z^2 - 6z\cos\frac{2\pi}{5} + 9 = 0 \text{ as req.}$$

29 d)

$$z_C = 3\,\mathrm{cis}\,\frac{4\pi}{5} \quad \text{and} \quad z_D = 3\,\mathrm{cis}\left(-\frac{4\pi}{5}\right).$$

Summing z_C and z_D:

$$\begin{aligned}
&3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right) + 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right) \\
&= 3\left[\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) - \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right] \\
&= 3\left[\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\frac{4\pi}{5} - \mathrm{i}\sin\frac{4\pi}{5}\right] = 3 \times 2\cos\frac{4\pi}{5}.
\end{aligned}$$

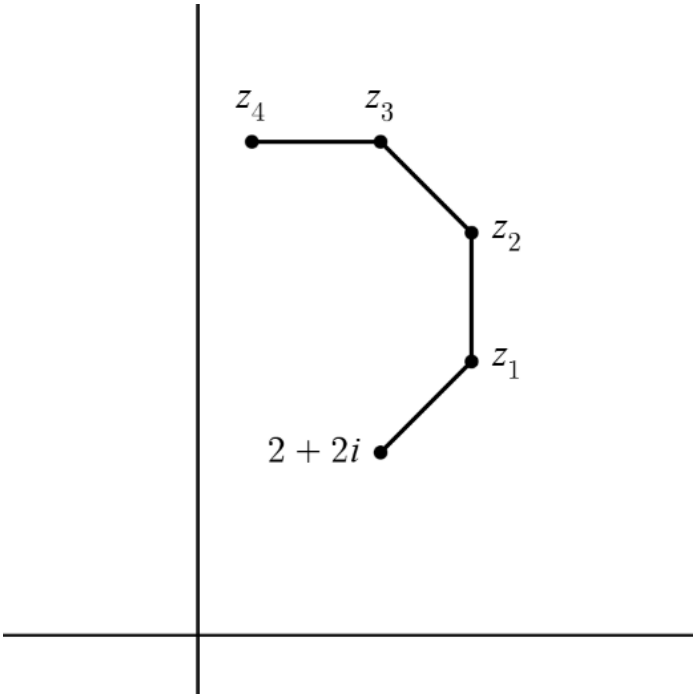
Multiplying z_C and z_D:

$$\begin{aligned}
&3\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right) 3\left(\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)\right) \\
&= 9\left(\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}\right)\left(\cos\frac{4\pi}{5} - \mathrm{i}\sin\frac{4\pi}{5}\right) \\
&= 9\left(\cos^2\frac{4\pi}{5} + \sin^2\frac{4\pi}{5}\right) = 9.
\end{aligned}$$

So that

$$z^2 - 6z\cos\frac{4\pi}{5} + 9 = 0.$$

30)



Set the problem to start at the origin and consider the vectors:

$$v_1 = \sqrt{2}e^{\pi i/4} = 1 + \mathrm{i}, \quad v_2 = \sqrt{2}e^{\pi i/2} = \sqrt{2}\mathrm{i}, \quad v_3 = \sqrt{2}e^{3\pi i/4} = -1 + \mathrm{i}, \quad v_4 = \sqrt{2}e^{\pi i} = -\sqrt{2}.$$

The sum:

$$1 + \mathrm{i} + \sqrt{2}\mathrm{i} - 1 + \mathrm{i} - \sqrt{2} = -\sqrt{2} + (2 + \sqrt{2})\mathrm{i}.$$

Now consider the displacement from $2 + 2\mathrm{i}$:

$$2 + 2\mathrm{i} - \sqrt{2} + (2 + \sqrt{2})\mathrm{i} = 2 - \sqrt{2} + (4 + \sqrt{2})\mathrm{i}.$$

31 a)

Solve $w^2 = 8 - 8\sqrt{3}\mathrm{i}$ where $w = z - 1$.

$$8 - 8\sqrt{3}\mathrm{i} = 16\left(\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right).$$

$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta).$$

$$r^2 = 16 \Rightarrow r = 4.$$

Since $2\theta = -\frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = -\frac{\pi}{6} + \frac{2k\pi}{2}.$$

$$k = 0: \quad z = 4\,\mathrm{cis}\left(-\frac{\pi}{6}\right) = 2\sqrt{3} - 2\mathrm{i}$$

$$k = 1: \quad z = 4\,\mathrm{cis}\,\frac{5\pi}{6} = -2\sqrt{3} + 2\mathrm{i}.$$

So $w = 2\sqrt{3} - 2\mathrm{i}$ and $w = -2\sqrt{3} + 2\mathrm{i}$ and so $z = w + 1$.

$$\therefore z = 1 + 2\sqrt{3} - 2\mathrm{i} \quad \text{and} \quad z = 1 - 2\sqrt{3} + 2\mathrm{i}.$$

31 b)

$w = z - 1$ so $|w|^2 = |z-1|^2$ which gives

$$|2\sqrt{3} - 2\mathrm{i}|^2 = 16 \quad \text{and} \quad |-2\sqrt{3} + 2\mathrm{i}|^2 = 16 \text{ units.}$$

For both roots it is 16 units.

31 c)

32 units.

31 d)

$$\begin{aligned} |z_1|^2 + |z_2|^2 &= |2\sqrt{3} - 2\mathrm{i}|^2 + |-2\sqrt{3} + 2\mathrm{i}|^2 \\ &= 17 + 4\sqrt{3} + 17 - 4\sqrt{3} = 34 \text{ units.} \end{aligned}$$

Alternatively:

$$32 + 1 + 1 = 34 \text{ units.}$$

32 a)

Solve $w^2 = -40 + 40\sqrt{3}\mathrm{i}$ where $w = z - 3$.

$$40 - 40\sqrt{3}\mathrm{i} = 80\left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right).$$

$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta).$$

$$r^2 = 80 \Rightarrow r = \sqrt{80}.$$

Since $2\theta = \frac{2\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{2\pi}{6} + \frac{2k\pi}{2}.$$

$$\begin{aligned} k = 0: \quad z_0 &= \sqrt{80}\,\mathrm{cis}\,\frac{2\pi}{3} = -2\sqrt{5} + 2\sqrt{15}\mathrm{i} \\ k = 1: \quad z_1 &= \sqrt{80}\,\mathrm{cis}\,\frac{4\pi}{3} = -2\sqrt{5} - 2\sqrt{15}\mathrm{i}. \end{aligned}$$

So $w = -2\sqrt{5} + 2\sqrt{15}\mathrm{i}$ and $w = -2\sqrt{5} - 2\sqrt{15}\mathrm{i}$ and so $z = w + 2$.

$$\therefore z = 3 + 2\sqrt{5} - 2\sqrt{15}\mathrm{i} \quad \text{and} \quad z = 3 + 2\sqrt{5} + 2\sqrt{15}\mathrm{i}.$$

32 b)

$$r = \sqrt{80} \text{ so } |z-3|^2 = 80 \text{ units.}$$

32 c)

160 units.

32 d)

$$|z_1|^2 + |z_2|^2 = |3 + 2\sqrt{5} + 2\sqrt{15}\mathrm{i}|^2 + |3 - 2\sqrt{5} - 2\sqrt{15}\mathrm{i}|^2$$
$$= 89 + 12\sqrt{5} + 89 - 12\sqrt{5} = 178 \text{ units.}$$

Alternatively:

$$160 + 9 + 9 = 178 \text{ units.}$$

33)

$$|z-2|^2 = (z-2)(z-2)^* = (z-2)(z^*-2) = zz^* - 2z - 2z^* + 4$$
$$= |z|^2 - 2(z + z^*) + 4.$$

If ω is any complex nth roots of unity then $|\omega|^2 = 1$.

$$\text{So that } |\omega - 2|^2 = 1 - 2(\omega + \omega^*) + 4 = 5 - 2(\omega + \omega^*) \text{ as req.}$$

34 a)

$$|z-3|^2 = (z-3)(z-3)^* = (z-3)(z^*-3) = zz^* - 3z - 3z^* + 9$$
$$= |z|^2 - 3(z + z^*) + 9.$$

If ω is any complex nth roots of unity then $|\omega|^2 = 1$.

$$\text{So that } |\omega - 3|^2 = 1 - 3(\omega + \omega^*) + 9 = 10 - 3(\omega + \omega^*) \text{ as req.}$$

34 b)

$$\sum_{k=0}^{4} |3 - \omega_k|^2 = \sum_{k=0}^{4} (10 - 3(\omega_k + (\omega_k)^*)) = 50 - 3\sum_{k=0}^{4} (\omega_k + (\omega_k)^*).$$

Since

$$\sum_{k=0}^{4} \omega_k = 0 \quad \text{and} \quad \sum_{k=0}^{4} (\omega_k)^* = 0,$$

$$\therefore (PA)^2 + (PB)^2 + (PC)^2 + (PD)^2 + (PE)^2 = 50.$$

35 a)

Since $3\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{3} \text{ which gives } ae^{0i} = a, \quad ae^{2\pi i/3} \quad \text{and} \quad ae^{-2\pi i/3}.$$

35 b)

$$a(1+\omega+\omega^2) = a \times 0 = 0.$$

35 c)

$$M_{AB} = \frac{A+B}{2} = \frac{a(1+\omega)}{2},$$
$$M_{BC} = \frac{B+C}{2} = \frac{a(\omega+\omega^2)}{2},$$
$$M_{CA} = \frac{C+A}{2} = \frac{a(\omega^2+1)}{2}.$$

Since $1+\omega+\omega^2 = 0$:

$$M_{AB} = -\frac{1}{2}a\omega^2, \quad M_{BC} = -\frac{1}{2}a, \quad M_{CA} = \frac{C+A}{2} = -\frac{1}{2}a\omega.$$

Which are the three cube-roots of unity scaled by $-\frac{1}{2}a$.

$$\begin{aligned}(z-p)(z-p\omega)(z-p\omega^2) &= (z^2 - p(1+\omega)z + p^2\omega)(z-p\omega^2) \\ &= z^3 - p(1+\omega)z^2 + p^2\omega z - p\omega^2 z^2 + p^2(\omega^2+\omega^3)z - p^3\omega^3.\end{aligned}$$

Since $\omega^3 = 1$:

$$\begin{aligned}&= z^3 - p(1+\omega)z^2 + p^2\omega z - p\omega^2 z^2 + p^2(\omega^2+1)z - p^3 \\ &= z^3 - p(1+\omega+\omega^2)z^2 + p^2(\omega^2+\omega+1)z - p^3 \\ &= z^3 - p^3.\end{aligned}$$

In this case, $p = -\frac{1}{2}a$.

$$\text{And so } z^3 - p^3 = z^3 - \left(-\frac{1}{2}a\right)^3 = z^3 + \frac{a^3}{8}.$$
$$\therefore 8z^3 + a^3 = 0.$$

36 a)

Since $5\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{5} \text{ which gives: } ae^{0i}, \quad ae^{2\pi i/5}, \quad ae^{4\pi i/5}, \quad ae^{6\pi i/5}, \quad ae^{8\pi i/5}.$$

$$ae^{0i}\ a, \quad ae^{2\pi i/5}, \quad ae^{4\pi i/5}, \quad ae^{-4\pi i/5} \quad \text{and} \quad ae^{-2\pi i/5}.$$

36 b)

$$a(1+\omega+\omega^2+\omega^3+\omega^4) = a \times 0 = 0.$$

36 c)

Since $e^{i\theta} = \cos\theta + \mathrm{i}\sin\theta$ then

$$\cos x = \frac{1}{2}(e^{ix} + e^{-ix}).$$

$$\begin{aligned}
\cos\left(\frac{\alpha-\beta}{2}\right) &= \frac{1}{2}\left(e^{(\alpha-\beta)i/2} + e^{-(\alpha-\beta)i/2}\right) \\
\Rightarrow 2\cos\left(\frac{\alpha-\beta}{2}\right) &= e^{(\alpha-\beta)i/2} + e^{-(\alpha-\beta)i/2} \\
\therefore 2\cos\left(\frac{\alpha-\beta}{2}\right)e^{(\alpha+\beta)i/2} &= e^{(\alpha+\beta)i/2}\left(e^{(\alpha-\beta)i/2} + e^{-(\alpha-\beta)i/2}\right) \\
&= e^{(\alpha+\beta)i/2}e^{(\alpha-\beta)i/2} + e^{(\alpha+\beta)i/2}e^{-(\alpha-\beta)i/2} \\
&= e^{2\alpha i/2} + e^{2\beta i/2} = e^{\alpha i} + e^{\beta i} \text{ as req.}
\end{aligned}$$

36 d)

A general midpoint is given by

$$\frac{a\omega^k + a\omega^{k+1}}{2}.$$

$$M_{AB} = \frac{A+B}{2}, \quad M_{BC} = \frac{B+C}{2}, \quad \text{etc}$$

From part c):

$$M_{AB} = \frac{ae^{\alpha i} + ae^{\beta i}}{2} = a\cos\left(\frac{\alpha-\beta}{2}\right)e^{(\alpha+\beta)i/2}.$$

When $\alpha = 0,\ \beta = \frac{2\pi}{5}$, then

$$M_{AB} = a\cos\frac{\pi}{5}e^{\pi i/5}.$$

Similarly for the other midpoints. The original roots satisfy $z^5 - a^5 = 0$. There is a scale factor of $\cos\frac{\pi}{5}$ so for any midpoint, w, then $w = z\cos\frac{\pi}{5}$.

$$z^5 - a^5 = 0 \text{ gives } \left(\frac{w}{\cos\frac{\pi}{5}}\right)^5 - a^5 = 0.$$

$$\therefore w^5 - \cos^5\left(\frac{\pi}{5}\right)a^5 = 0 \text{ as req.}$$

36 e)

$$w^6 - \cos^6\left(\frac{\pi}{6}\right)a^6 = 0 \Rightarrow w^6 - \frac{27}{64}a^6 = 0.$$

$$\therefore 64w^6 - 27a^6 = 0.$$